

# Implementing Logistic Regression to Evaluate Cardiac condition and Time Series forecast on Air temperature dataset

Jacob Benny Packiaraj  
M.Sc in Data Analytics  
National College of Ireland  
Dublin,Ireland  
x22188801@student.ncirl.ie

**Abstract**— Assessing the cardiac condition and building a statistical model for classification is an important step to identify the detection of accurate cases. In this paper, the Logistic Regression model is applied on the cardiac condition dataset and the performance on the model is evaluated with the test data. F1 scores, Precision and Recall accuracy and balanced accuracy score is derived and is checked for the final model. This paper also implements a time series forecasting on weather dataset on how the parameter- maximum air temperature in degree Celsius, will be forecasted using different time series technique and identify the model that closely forecast the temperature based on the underlying patterns between the year 2019 and 2022 to forecast the temperature for 2023. A logistic model is built with an F1 score of 0.62 and balanced accuracy score of 0.726. For the timeseries analysis, Simple , exponential and ARIMA models were forecasted and the weighted moving average shows the lowest residuals .

**Keywords**—*Correlation, ARIMA, SARIMA, Auto Correlation function, Partial autocorrelation function. component,*

## I. INTRODUCTION

Classification of cases whether the case has a cardiac condition or not is a critical task and having a machine learning algorithm to identify the condition based on the regressors like age, gender, weight, and other factors influence the condition. This may be due to presence of a linear association between the independent variables to the dependent variables. One such algorithm used is the logistic regression to do classification of cases where the association between the independent variables and dependent variables is used to predict the cardiac condition of a test data.

Time series is a vast area in statistics to evaluate and understand the data that is collected over time rather than the cross-sectional data. There are algorithms present based on the underlying pattern in the data to accurately forecast the future values of a parameter. In this paper, weather data of Ireland from 1942 till 2023 is available where the maximum temperature parameter is forecasted based on training the different timeseries model using the data for period 2019 to 2022.

## II. DATA EXPLORATION

The implementation of this report deals with two datasets – one for the classification technique- Logistic Regression and the other for Time series forecasting weather dataset.

### A. Cardiac dataset

The dataset has six features, out of which five are relevant to be used in the regression model. Case number is

dropped from the dataset as it is a unique identifier in the dataframe.

- Numeric Data:

Age column in the dataset represent the age of the person which is an integer. Weight feature is the weight of the person that has a floating data type. The final numerical feature is the fitness score of the case. This has a float data type and has data for all the individual observations in the dataset.

- Categorical data

The gender of the observation is given as a categorical value as Male or Female which is one of the independent variables.

The final categorical data is the dependent variable – Cardiac condition which is a bivariate data that has the value either Present or Absent.

The unique identifier of the observations in the dataset case number is not used for modelling the regression and is dropped from the model.

### B. Weather dataset

For the time series analysis, Weather data from 1942 to 2023 is used.

- Date Time

The unique identifier of this dataset is the date field which has the data type datetime to identify the individual observations in the dataset.

- Numerical data

The numerical dataset available in the dataset are the maximum air temperature in degree Celsius which is the temperature of the air recorded every day in Celsius measurement.

Similarly minimum air temperature is recorded for the individual observation in degree Celsius. The final temperature specific feature in the data is the Grass minimum temperature.

All the temperature data are in floating point values.

The other features available are Mean cbl pressure which is the atmospheric pressure recorded on any day. This is measured in Hectopascal(hPa) and it is of floating datatype.

Mean Windspeed at the time of observation is captured in the dataset and it has the unit of measurement in knots.

Potential Evapotranspiration is the possible evaporation of water to the atmosphere from the surface which is measured in milli meter.

The final parameter is Evaporation which is the quantity at which the water is evaporated to the atmosphere. This feature is measured in milli meter.

Most of the data in this dataset has the data type of a floating point of measurement. Out of all the features available in the dataset, this paper deals with the forecasting of the feature- Maximum air temperature in degree Celsius.

### III. DESCRIPTIVE STATISTICS

#### A. Logistic regression

The data from the cardiac dataset is checked for presence of any null values or NA values.

To understand the linear relationship between the independent and dependent variables, scatterplot is plotted.

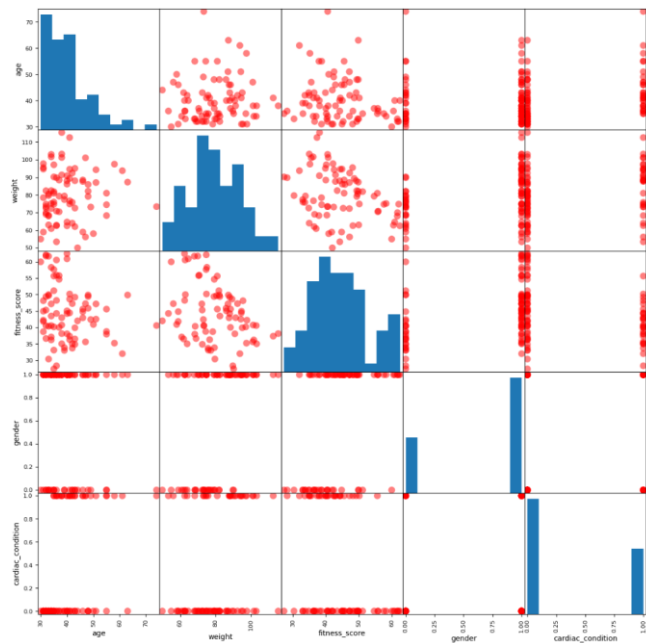


Fig. 1. Scatterplot between variables for logistic regression

It is observed that, there is a random pattern for the columns among age, weight, and fitness\_score and there is no linear relationship between them to cause collinearity. Similarly, for the categorical data of gender and cardiac condition, the scatter plot is observed for detecting any patterns.

To confirm there is no linear relationship among the independent variables, correlation is calculated for these variables and is plotted as a heatmap. From the correlation values calculated using Pearson method, there is no serious correlation among the independent variables.

The data is verified for the presence of any outliers, and it can be observed that Age feature has some outliers. In real time, people in older age tend to get cardiac condition. Due to this reason, the outliers are not removed from modelling the regression.

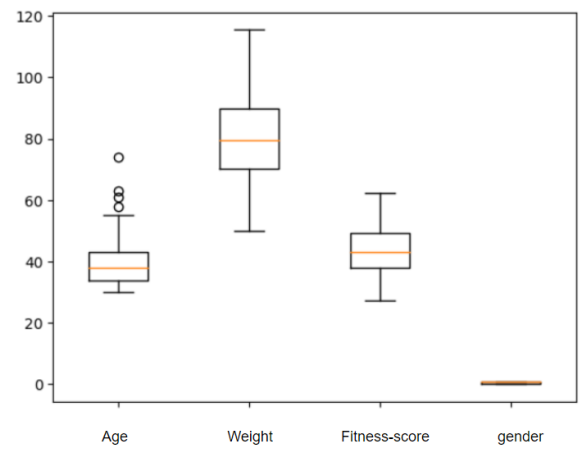


Fig. 2. Boxplot for identifying outliers in cardiac dataset

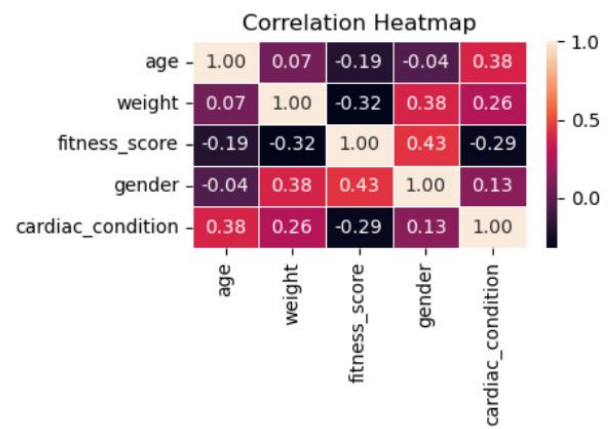


Fig. 3. Correlation heatmap for regression analysis

To check for any influential point or outlier in the dataset, box plot is plotted for the training independent variables and could see there are 4 observations are present in the training dataset for the Age feature.

Regplot for the logistic estimate of the regression model for the individual feature is plotted against the dependent variables.

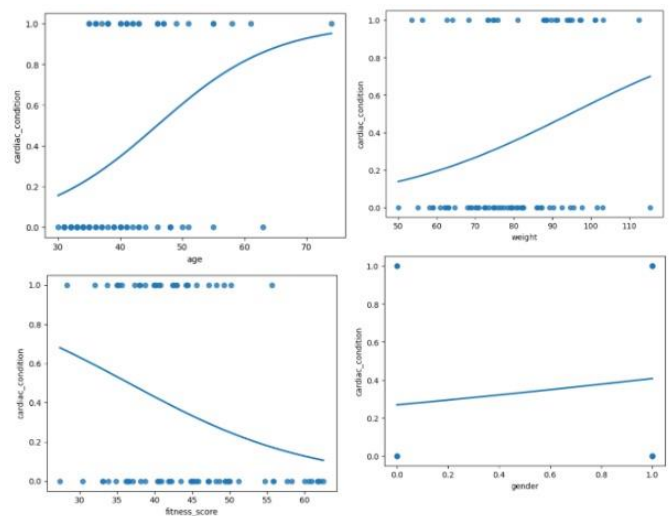


Fig. 4. Regplot on the independent vs dependent variables

It is inferred from the plot that people having lower age also have cardiac condition. Similarly for the weight and fitness\_score, case having cardiac condition is distributed for all the parameter in the group.

### B. Time Series Forecasting

The exploratory analysis of the maximum air temperature field is carried out in python and the underlying pattern of the data is analysed for the temperature observation between 2019 and 2022.

The first step of the exploration is to plot the time series data of temperature against time to have a view of the nature of the data and its actual pattern. It is observed that the temperature for the four years follows a uniform pattern showing a seasonal trend where the temperature reaches a maximum during the month of July and reaches lowest at around January.

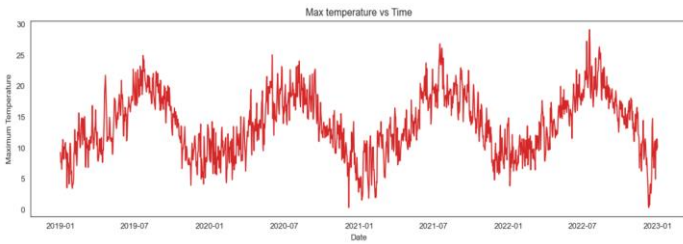


Fig. 5. Visual analysis of maximum air temperature

Plotting of a histogram with the temperature data is carried out and the temperature observation is normally distributed.

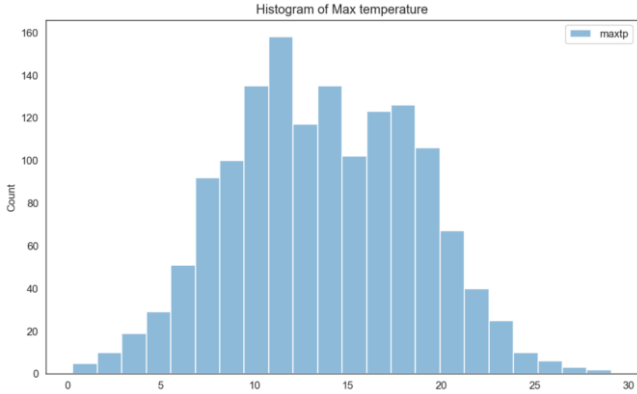


Fig. 6. Histogram on maximum air temperature

During the analysis, outliers are verified in the sample dataset and from the boxplot, an notifiable observation is present in the maximum air temperature. This is an influential observation and not an outlier and hence it is not removed from the modelling of the algorithm.

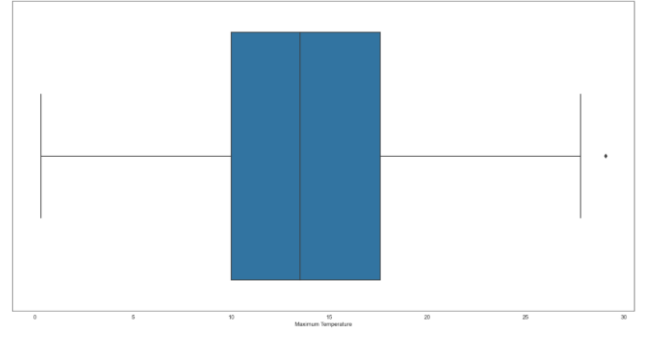


Fig. 7. Box plot for validating outliers

## IV. MODELLING PROCESS

### A. Logistic Regression

Data transformation is applied to the columns -gender and cardiac condition as it has categorical data, and the python model requires a numeric data sent to predict the model. getDummies function is used from the Pandas library to transform the bivariate data into zeroes and ones.

The cardiac dataset has 100 observations and is split into train and test dataset using the student id as the seed. The train data is used to fit the model and test data to evaluate the model.

In this paper, Logistic Regression method from statsmodel is used to fit and evaluate the model. The train data having the independent variables and the object having the dependent variables is passed to the Logit function of the Scikit module statsmodel. This uses the maximum likelihood estimation, the method of estimating the parameters of an assumed probability distribution, provided some observed data.

On printing the fit summary, the regression result is printed in the notebook.

The logistic regression has a hypothesis testing where the null hypothesis is that all the coefficients in the model are equal to zero. The alternative hypothesis is that not every coefficient is simultaneously equal to zero.

To find the individual regressors being significant to the model, individual chi square test is conducted to evaluate the parameter being significant to the model.

The summary statistic of the individual regressors shows that the feature Weight has probability greater than 0.05 and hence it is insignificant to the model. Weight feature is dropped from the model and the model is again fitted with the reduced feature.

All the features have statistically significant scores of probabilities less than 0.05 and this model is the fitted model taken for evaluation.

$$E(y) = \frac{e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}}$$

To get the log odds of the fitted model, exponential of the coefficients calculates is taken.

The odds ratio for a variable represents how the odds change with a unit increase in that variable holding other variables constant.

The interpretation of the log odds from the fitted model is that the odds of getting a cardiac condition for an older person is 1.11 times the odds of a young person. Or the older person has 11% more odds of having a cardiac condition compared to normal person.

Similarly, the odd of having a cardiac condition is 0.85 times the odds of a person having poor fitness score. In other words, person having higher fitness score has 15% less odds of having a cardiac condition.

The odds of having a cardiac condition for a male is 8.32 times that of a female.

To verify the significance of the features relevant to be statistically significant to the model, Wald test is carried out. For the features Age, Fitness score and gender, the wald test function is invoked on the fitted model. A test statistic of 12.73 is measured and the probability of it is close to 0 which is less than 0.05. Hence the parameters are statistically significant to the model.

### B. Time Series forecasting

In forecasting the maximum air temperature, to find the suitable model, data between 2019 and 2022 is used for training purpose and this value is used for the prediction of the temperature estimates in 2023.

The data is read from the csv file and is stored in a dataframe. Further check is done to find the presence of any null values or missing values in the dataset. The dataframe is split into the train and test data for the mentioned time range. The transformation of the datetime field is applied to be read as a numpy datetime field.

This paper deals with the estimation of the evaluation of simple timeseries, model building using exponential smoothing and implementation of ARIMA/SARIMA model.

#### 1. Simple timeseries model:

- Mean method

One of the simplest timeseries forecast is the mean method, where the cumulative mean of the daily actual temperature is the future forecast temperature. This method is not the actual fit as the past data from a farthest time is used which is not significant for forecasting.

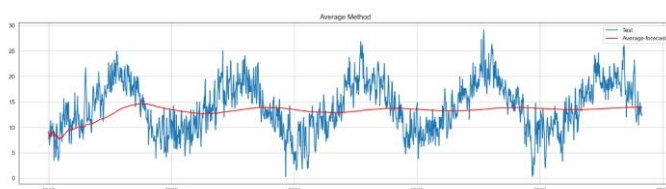


Fig. 8. Mean method forecast

The MSE calculated is 24.36 and MAE is 4.076

- Naïve method

Naïve forecast is the easiest technique where the forecast of the next day temperature is the today's temperature. In other words, the forecasted temperature is a day lag of the present temperature.

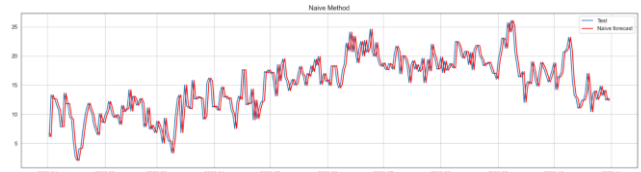


Fig. 9. Naïve method forecast

This model has the MSE of 4.9 and MAE of 1.70

- Simple moving average

In simple moving average, the weights given to the window of past temperature data is distributed equally. In this analysis, a rolling mean of 7, which is a 7-day moving average is calculated where the average of the past seven actual temperature will be the future forecasted temperature.

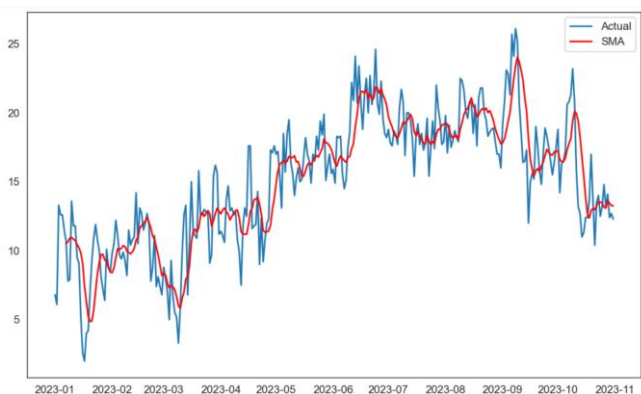


Fig. 10. 7-day simple moving average

Here, the noise in the actual data is removed due to the averaging of the temperature data and a smoother curve is obtained.

- Weighted Moving Average

In Weighted moving average, a 7-day rolling mean along with the weights associated with each temperature data with the recent data having more weightage associated and the past data has the least weightage associated to it.

In this paper an array of 7 weights [0.0078125, 0.015625, 0.03125, 0.0625, 0.125, 0.25, 0.5] is used to calculate the 7-day weighted moving average forecast.

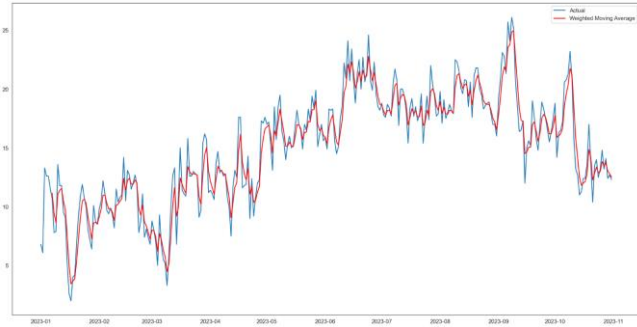


Fig. 11. 7-day weighted moving average

- Exponential smoothing

To proceed with performing exponential smoothing of the temperature data, decomposition of the actual data is performed to find the underlying pattern of the level, trend, seasonality and error present in the data.

Based on this, the variable can be decomposed either as an additive or multiplicative where all the underlying pattern are either the products or sums of the individual parameters.

After performing the decomposition of this univariate data, the pattern in figure is observed.

It is observed that in the multiplicative and additive decomposing of the maximum air temperature data, there is a seasonality component present in the data which is identified.

There is no sudden jump in the temperature data throughout the year as there is no multiplicative effect on the underlying component is present. Hence this time series model exhibits an additive model for the maximum air temperature.

$$\text{Additive } x_t = \text{Trend} + \text{Seasonal} + \text{Random}$$

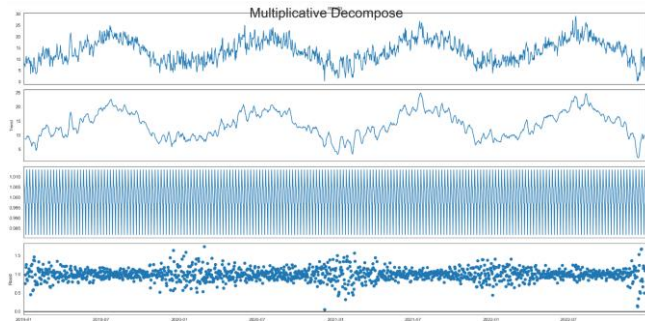


Fig. 12. Multiplicative decomposition of temperature data

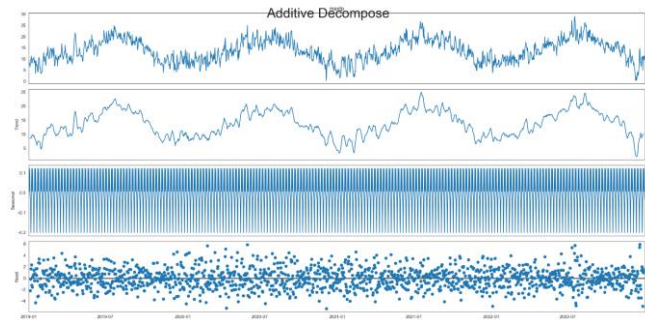


Fig. 13. Additive decomposition of temperature data

- Triple Exponential smoothing

Since the data has seasonality part, which is identified during the decomposition of the data, this paper implements the Holt Winter triple exponential smoothing for this univariate data. Holt-Winters is effective when the temperature exhibits both a trend and a seasonal component.

Stats model package is used to calculate the holt winter forecast of the data. The forecast of the 2023 data is calculated by fitting the data from 2019 to 2022 to the Exponential smoothing method is invoked from the statsmodel for the trend and seasonality showing additive and is for the seasonal period of 365. The model evaluated and provided the smoothing level, smoothing trend and smoothing seasonal as 0.63, 2.4e-9, 2.6e-8 as the coefficients.

The forecast and the actual data from the triple exponential method is plotted as below.

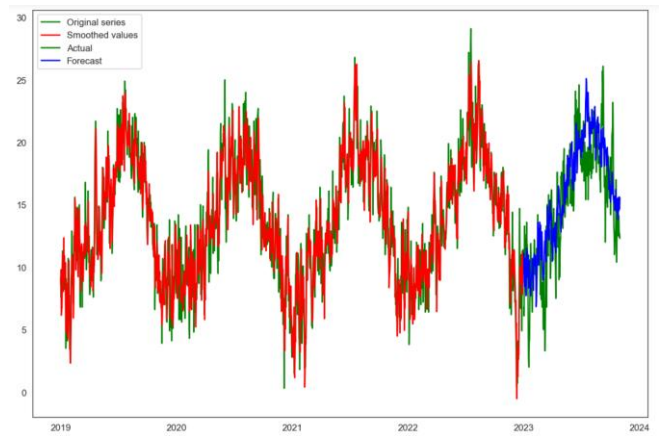


Fig. 14. Holt- Winter – triple exponential smoothing

- ARIMA Forecast

Stationarity of the temperature data is evaluated to check if the actual data is stationary or non-stationary meaning have a constant mean and variance in the time series.

Augmented dickey fuller test is evaluated for the training dataset and the probability value is greater than 0.05. This suggest the data is nonstationary and to construct a model, the data needs to be transformed to make it stationary.

The null hypothesis in the ADF test is that the data is stationary, and the alternative hypothesis is the data is non-stationary.

The first order differencing is the transformation technique used in this paper to make the data stationary.

ADF test is conducted again, and the probability statistic is in the significant zone saying the data is stationary.

Autocorrelation function (ACF) and Partial Autocorrelation function (PACF) is plotted for the training data and are used to figure out the order of the AR and MA for the AR and MA models.

As the raw data is nonstationary, first order differencing is carried out and hence the d variable in building the ARIMA model is 1 in this case.



By analyzing the ACF and PACF plot, the lags of the AR and MA model are identified to be 1,1. Hence an ARIMA model of (1,1,1).

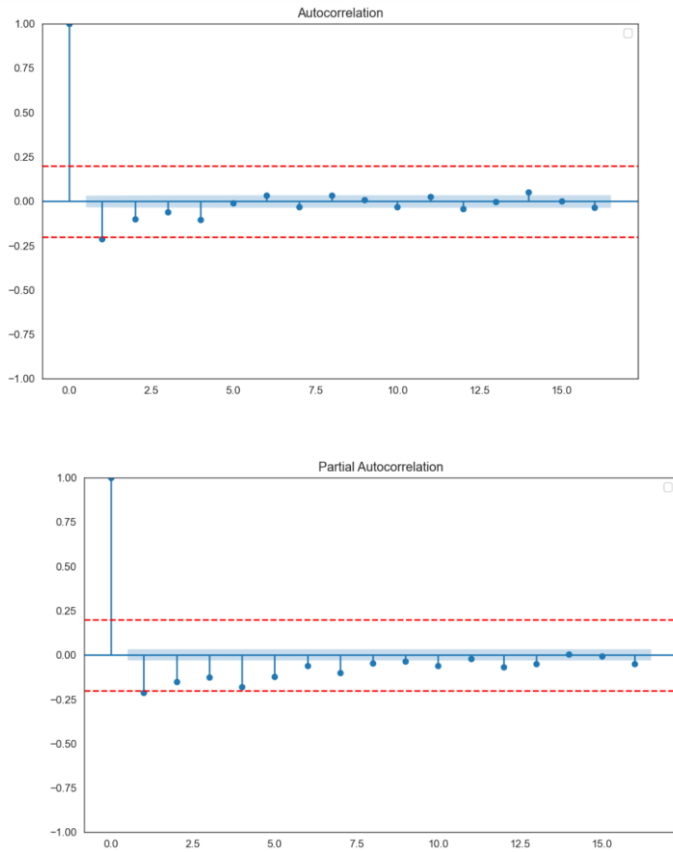


Fig. 15. ACF and PACF plots

Also, the other models for the other AR, MA values have AIC score higher than the (1,1,1) model and hence ARIMA (1,1,1) is chosen. The forecasted values is predicted from the statsmodels package and below plot is attained.

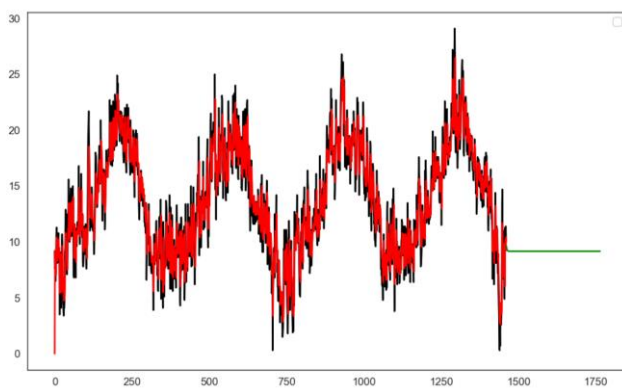


Fig. 16. ARIMA (1,1,1) forecast

The model is statistically significant, where the AR and MA component has the probability of less than 0.05.

- Seasonal ARIMA

Seasonal Auto Regressive Integrated Moving Average is chosen for the analysis as the data has the seasonal component.

An iterative procedure is carried out to identify the p,d and q value of the SARIMAX model and the model having the lowest AIC score is picked for the modeling and forecasting the temperature data.

The model with the components SARIMA(1, 1, 1)x(0, 1, 1, 12)12 has obtained the lowest AIC:7633.556 score. The model is fitted and built using the configurations.

|                                 |                          |
|---------------------------------|--------------------------|
| SARIMA(1, 0, 0)x(0, 0, 1, 12)12 | - AIC:7896.733121246569  |
| SARIMA(1, 0, 0)x(0, 1, 0, 12)12 | - AIC:8816.617504893937  |
| SARIMA(1, 0, 0)x(0, 1, 1, 12)12 | - AIC:7803.753693320416  |
| SARIMA(1, 0, 0)x(1, 0, 0, 12)12 | - AIC:7896.673856847001  |
| SARIMA(1, 0, 0)x(1, 0, 1, 12)12 | - AIC:7898.494861819239  |
| SARIMA(1, 0, 0)x(1, 1, 0, 12)12 | - AIC:8243.071605250207  |
| SARIMA(1, 0, 0)x(1, 1, 1, 12)12 | - AIC:7805.753568897471  |
| SARIMA(1, 0, 1)x(0, 0, 0, 12)12 | - AIC:7830.924377669025  |
| SARIMA(1, 0, 1)x(0, 0, 1, 12)12 | - AIC:7784.620122594481  |
| SARIMA(1, 0, 1)x(0, 1, 0, 12)12 | - AIC:8812.81647398668   |
| SARIMA(1, 0, 1)x(0, 1, 1, 12)12 | - AIC:7732.807581227518  |
| SARIMA(1, 0, 1)x(1, 0, 0, 12)12 | - AIC:7788.114979992998  |
| SARIMA(1, 0, 1)x(1, 0, 1, 12)12 | - AIC:7786.182343615598  |
| SARIMA(1, 0, 1)x(1, 1, 0, 12)12 | - AIC:8235.618333735209  |
| SARIMA(1, 0, 1)x(1, 1, 1, 12)12 | - AIC:7734.806171660399  |
| SARIMA(1, 1, 0)x(0, 0, 0, 12)12 | - AIC:7870.874310861995  |
| SARIMA(1, 1, 0)x(0, 0, 1, 12)12 | - AIC:7823.664858452594  |
| SARIMA(1, 1, 0)x(0, 1, 0, 12)12 | - AIC:9088.967882205287  |
| SARIMA(1, 1, 0)x(0, 1, 1, 12)12 | - AIC:7816.196230113679  |
| SARIMA(1, 1, 0)x(1, 0, 0, 12)12 | - AIC:7823.601207283164  |
| SARIMA(1, 1, 0)x(1, 0, 1, 12)12 | - AIC:7823.740707487997  |
| SARIMA(1, 1, 0)x(1, 1, 0, 12)12 | - AIC:8455.764041976514  |
| SARIMA(1, 1, 0)x(1, 1, 1, 12)12 | - AIC:7828.453659807456  |
| SARIMA(1, 1, 1)x(0, 0, 0, 12)12 | - AIC:7683.256280751437  |
| SARIMA(1, 1, 1)x(0, 0, 1, 12)12 | - AIC:7637.8178666314125 |
| SARIMA(1, 1, 1)x(0, 1, 0, 12)12 | - AIC:8816.825875955872  |
| SARIMA(1, 1, 1)x(0, 1, 1, 12)12 | - AIC:7633.556331867914  |
| SARIMA(1, 1, 1)x(1, 0, 0, 12)12 | - AIC:7641.8362392428135 |
| SARIMA(1, 1, 1)x(1, 0, 1, 12)12 | - AIC:7638.836098293781  |
| SARIMA(1, 1, 1)x(1, 1, 0, 12)12 | - AIC:8238.260595408001  |
| SARIMA(1, 1, 1)x(1, 1, 1, 12)12 | - AIC:7635.550458258769  |

Fig. 17. Least AIC score from Iterating for SARIMA

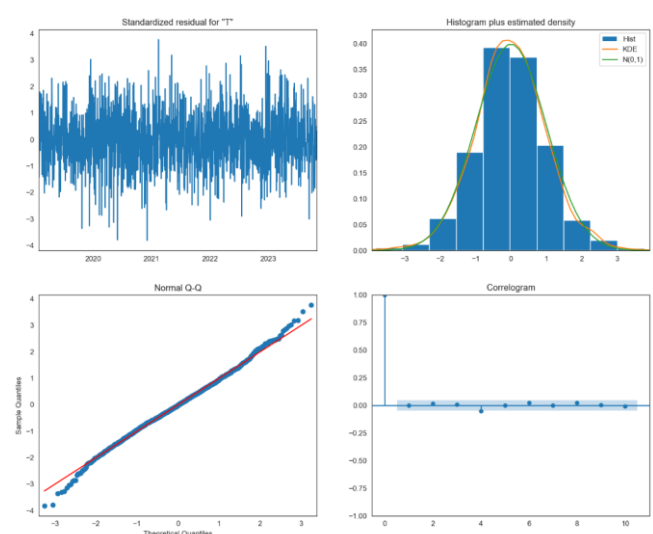


Fig. 18. Residual analysis of SARIMAX model

The forecasted value obtained from this model is evaluated and is plotted against the actual value from 2023.

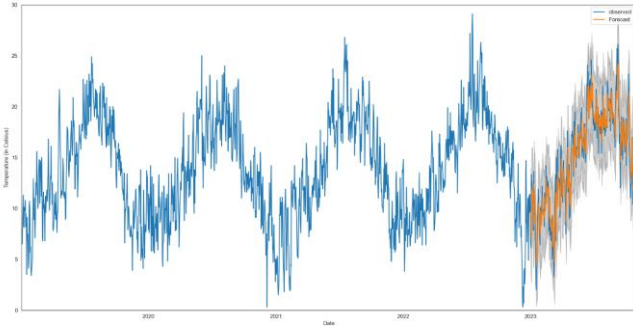


Fig. 19. SARIMAX forecast

The residuals are calculated and the MAE and MSE obtained for this model is 1.69 and 4.96 respectively.

## V. EVALUATION OF MODEL PARAMETERS

### A. TimeSeries Forecasting

To evaluate the goodness of the forecast of the time-series model, Mean Absolute Error (MAE), Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) is calculated to find the best model for the forecasting of this maximum air temperature data.

TABLE I. MODEL EVALUATION

| Forecasting method | Evaluation method |       |      |
|--------------------|-------------------|-------|------|
|                    | MAE               | MSE   | MAPE |
| Mean method        | 4.076897          | 24.36 | 45.7 |
| Naïve method       | 1.70132           | 4.9   | 13.2 |
| SMA                | 1.693432          | 4.96  | 14.3 |
| WMA                | 0.857023          | 1.19  | 6.88 |
| Holt Winter        | 2.424297          | 9.19  | 21.5 |
| ARIMA(1,1,1)       | 6.441415          | 57.13 | NaN  |
| SARIMA             | 1.620551          | 4.37  | 13.3 |

The difference between the Actual and forecasted point in the dataset for 2023 is used for the calculation of all the scores to find the best model for the forecasting.

From the forecasted values the average method has the Mean Absolute Error of 4.06, mean Squared Error of 24.36 and Mean Absolute Percentage Error of 45.7. The value is high for this method as the cumulative average is for the time when the temperature data is known. Here the average is calculated from 2019 for the forecast in 2023.

Naïve forecast has less error where the MAE is 1.7013, MSE – 4.9 and MAPE – 13.2. This forecast performs well as the future forecast will be today's data. Hence the error is significantly less.

For the 7-day Simple Moving Average, MAE is 1.69, MSE is 4.96 and MAPE is 14.3. Here the smoothing of the curve occurs showing a decrease in the noise in data.

Weighted Moving Average is the best fit for forecasting among the methods used. MAE of 0.85, MSE of 1.19 and MAPE of 6.88 is achieved. In this method, the wights close to 50 % are provided to the previous actual value and is decreased significantly to the other 6 days.

Since the data shows additive seasonal component in the data, Holt winter – triple exponential smoothing get the error of MAE- 2.434, MSE- 9.19 and MAPE- 21.5.

In the ARIMA and SARIMA models, ARIMA(1,1,1) also predicted by the auto-ARIMA function has the error of MAE- 6.44, MSE- 57.13. Similarly for SARIMA forecast- the MAE of 1.62, MSE of 4.37 and MAPE of 13.3 is calculated.

Of the forecasting models chosen, Weighted Moving average followed by SARIMA and by Naïve method are the better forecasting methods for predicting the maximum air temperature.

### B. Logistic Regression

Classification report of the regression model is printed to check on validating the accuracy of the model. Precision, recall and F1 score of the model is calculated.

Precision- Out of all the cases the model predicted, only 57% was detected with correct cardiac condition.

Recall- Out of all the cases that has cardiac condition, the model only predicted this outcome only for 67%.

The F1 score is calculated by the below formula.

$$F1\ score = 2 * \frac{(Precision * Recall)}{(Precision + Recall)}$$

The calculated F1 score is 0.62.

The balanced accuracy score for the fitted model to assess the performance of the classification model.

$$Balanced\ Accuracy = \frac{Sensitivity + Specificity}{2}$$

From the actual and the predicted value of the logistic regression, the balanced accuracy score obtained is 0.726.

Confusion matrix is plotted for the actual and predicted data and the distribution of the true positive, true negative, false positive and false negative is used to understand whether the model has detected the correct and false cases.

It is interpreted that the model has correctly predicted 4 cases as True positive and 11 cases having True negative. 2 cases have false negative which is danger in real time where the model doesn't detect the cases correctly when the cardiac condition is present. The model has identified 3 cases as false positive where the model has predicted the cardiac condition as yes, but the case has no actual cardiac condition in the first place.

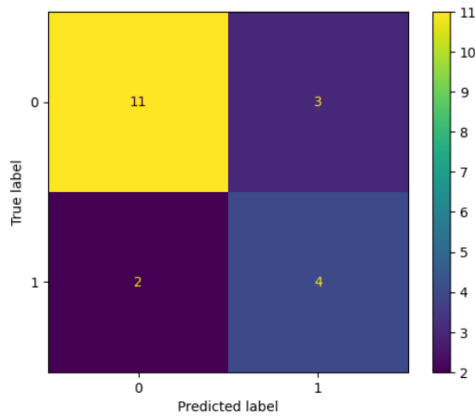
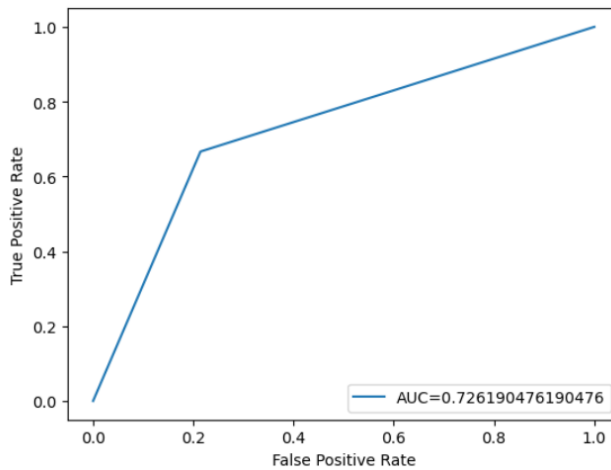


Fig. 20. Confusion Matrix of the test data



AUC-ROC curve is plotted with true positive rate against false positive rate. This is plotted to measure how much the model is capable of distinguishing between the classes. The AUC obtained is 0.726. There is 12% predictive capacity in the logistic model compared no skill classifier model.

PRC curve for precision recall parameter is plotted to evaluate the sensitivity of the model. However, the model

built has low accuracy and parameter tuning needs to be carried out to increase the accuracy of the prediction capacity of the model.

## VI. CONCLUSION

For the logistic regression model, 15 of the 20 cases were predicted correctly by the model and for 2 cases, the model has wrongly predicted the condition where the underlying case does have the cardiac condition.

The model is having moderate accuracy in classifying the cases and it can be improved by parameter tuning or other transformation.

For the time series data, the weighted moving average having the weights as discussed in the paper, closely forecasted the data by minimizing the errors. Similarly, SARIMA forecast is next in the line with the minimized residual errors.